Retractability problem for quasi-linear cycle sets

Ilaria Colazzo
(University of Exeter)

Plan

1. Basic definitions
2. Quasi-linear cycle sets & Rump's conjecture
3. An idea of a proof for Rump' conjecture (joint work, still in progress with L. Vendramin)

1 Basic definitions

Def A pair (X,r) is a (set-theoretical) solution to the YBE if X is a set and $r: X\times X \to X\times X$ s.t.

$$(r\times id)(id\times r)(r\times id) = (id\times r)(r\times id)(id\times r)$$

We write

$$r(x,y) = (\lambda_x(y), \rho_y(x))$$

where $\lambda_x, \rho_x : X \to X$ are maps.

Def (X,r) a s.t. solution to the YBE we say that (X,r) is <u>involutive</u> if $r^2 = id_{X\times X}$ and is <u>non-degenerate</u> if λ_x, ρ_x are bijective for any $x\in X$.

<u>Convention</u> When I say <u>solution</u> I mean

a set-theoretical solution which is involutive and non-degenerate.

<u>Examples</u>

(1) Let X be a group. The flip map $r(x,y)=(y,x)$ is a solution.

(2) Let X be a set and $f, g \in \mathrm{Sym}\, X$ the map
$$r(x,y)=(f(y), g(x))$$
is a solution if and only if $fg=gf$.
(X,r) is involutive iff $g=f^{-1}$.
This solutions are called <u>permutation solutions</u>

<u>Def</u> Let (X,r) be a solution we define the following equivalent relation
$$x \sim y \iff \lambda_x = \lambda_y.$$

<u>Fact</u> $\sim$ iduces a solution on the $X/\sim$ called the retraction of (X,r) and denoted by $\mathrm{Ret}(X,r)$.

<u>Def</u>[ESS99] (X,r) is a solution we define
$$\mathrm{Ret}^1(X,r) = \mathrm{Ret}(X,r)$$
$k>1$ $\mathrm{Ret}^k(X,r) = \mathrm{Ret}(\mathrm{Ret}^{k-1}(X,r))$
We say that (X,r) is a <u>multipermutation solution</u> if there exists $m \in \mathbb{N}$ s.t. $|\mathrm{Ret}^m(X,r)|=1$

<u>Conjecture</u>[GI04] All finite <u>square-free</u> solutions ($r(x,x)=(x,x)$) (X,r) with $|X| \geq 2$ are multipermutation solutions.

This conjecture is false [Ve16].

Rump in 2016 introduced a family of solutions which are not square-free.

2. Quasi-linear cycle sets & Rump's conjecture

<u>Def</u> [Ru05] A cycle set is a $(X, \cdot)$ where X is a set and $\cdot$ is binary operation that satisfies

$$(x \cdot y) \cdot (x \cdot z) = (y \cdot x) \cdot (y \cdot z) \qquad \text{(CS)}$$

for all $x, y, z \in X$. and the left traslations $\sigma_x : y \mapsto x \cdot y$ are bijective

<u>Fact</u> finite cycl sets $\longleftrightarrow$ solutions (involutive and non deg.)

$$(X, \cdot) \longmapsto r(x,y) = (\sigma_x^{-1}(y), \sigma_{\sigma_x^{-1}(y)}(x))$$

$$(X, \cdot) \text{ where } x \cdot y = \lambda_x^{-1}(y) \longleftarrow r(x,y) = (\lambda_x(y), \rho_y(x))$$

<u>Def</u> An abelian group $(A,+)$ with a bijective map $\tau : A \to A$ s.t $\tau(0) = 0$ is called a <u>τ-group</u>. A τ-group (A, τ) s.t. the binary operation defined by

$$a \cdot b = \tau(b-a) - \tau(-a)$$

turns A into a cycle set is called a <u>quasi linear cycle set</u>.

<u>Fact</u> (A, r) a r-group. with $a \cdot b = r(b-a) - r(-a)$
the TFAE

(1) $(A, \cdot)$ is a cycle set

(2) $\forall a, b \in A \quad r(a \cdot b) = r(a) \cdot r(b)$

TABLE 1.1. Some values of $q(n)$.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$q(n)$	1	1	2	5	4	6	6	45	20	14	10	115	12	29	36

<u>Conjecture 1</u> If $p \in \mathbb{P}$. For any quasi-linear cycle set on A of cardinality p then the $r \in \mathrm{Aut}(A, +)$.

If conjecture 1 is true this means that if (A, r) is a q.l cycleset with $|A| = p \in \mathbb{P}$ the $a \cdot b = r(b)$ in the associated solution we have that $\lambda_x = r^{-1} \ \forall x \in X$.

Rump tested the conjecture for < 13

I " " " for < 23

<u>Conjecture 2</u> The solution assiated to a quasi-linear cycle set is always multipermutational.

Computer evidence for $|X| < 23$.

Conj. 2 $\Rightarrow$ conj 1.

3. <u>An idea of a proof of conjecture 1</u>

- $p \in \mathbb{P}$
- $(\mathbb{Z}/p, r)$ a quasi-linear cycle set

(*) $\quad a \cdot b = r(b-a) - r(-a)$

this is equivalent to

(1) $\tau(0) = 0$

(2) $\tau(a \cdot b) = \tau(a) \cdot \tau(b)$

Write (2) without using $\cdot$

(3) $\tau(\tau(b-a) - \tau(-a)) = \tau(\tau(b) - \tau(a)) - \tau(-\tau(a))$

We want to prove that $\tau \in \mathrm{Aut}(A, +)$.

<u>Idea</u> Rewrite τ as a polynomial on the field of p elements

$$T(X) = \sum_{a \in \mathbb{Z}/p} \tau(a)\left(1 - (X-a)^{p-1}\right)$$

$\deg T \leq p-2$ because the coefficient of X^{p-1} is $\sum_{a \in \mathbb{Z}/p} a = 0$. Moreover $T(0) = 0$.

Rewrite (3) as a polynomial identity

(3') $T(T(X-Y) - T(-Y)) = T(T(X) - T(Y)) - T(-T(Y))$

Assume (3') is true.

Claim that $T'(0) \neq 0$. If not take the derivative of (3') wrt Y and

$$T'(T(X-Y) - T(-Y))(-T'(X-Y) + T'(Y)) =$$
$$= T'(T(X) - T(Y))(-T'(Y) + T'(-T(Y))(T'(Y))$$

for $Y = 0$

$$T'(T(X))\, T'(X) = 0$$

$\rightarrow T'(X) = 0$

$\rightarrow T(X) = 0$ contradiction.

Apply the derivative wrt X to (3')

$$T'(T(X-Y) - T(-Y))(T'(X-Y)) =$$
$$= T'(T(X) - T(Y))\,T'(X)$$

for $X=Y$

$$T'(-T(-X))\,T'(0) = T'(0)\,T'(X)$$

$\rightarrow$ $T'(-T(-X)) = T'(X)$

Now if $d = \deg T(X)$

$$d(d-1) = (d-1)$$

$\rightarrow$ $d = 1$.

$\rightarrow$ $\tau \in \mathrm{Aut}(A|t)$.